

Homework

Basics in linear algebra and numerical methods, 2025/26. semester I.

General instructions: Each week, you have to submit the solution of three arbitrary exercises. This should be done as a Python code. Comments should be written in English.

Scores: 3×3 per week; altogether: 54 in the whole semester.

Week 1

- (a) Take vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^{10}$ and build $A \in \mathbb{R}^{20 \times 20}$ such that its main diagonal is $(u_1, v_1, u_2, v_2, \dots, u_{10}, v_{10})$ and their off diagonals are $(v_{10}, u_{10}, v_9, u_9, \dots, v_2, u_2, v_1)$. If possible, use vectorized operations.
- (b) Take a random vector $\mathbf{v} \in \mathbb{R}^{20}$ with values between -3 and 3. Then, remove those elements, which are in $[-1, 1]$. You may also use logical "**and**".
- (c) Take a random vector of length 100 and compute the vector $\mathbf{v}_{\text{new}}$, which is the difference of the neighboring elements with $\mathbf{v}_{\text{new}}[j] = \mathbf{v}[j] - \mathbf{v}[j - 1]$ for $j > 0$ and $\mathbf{v}_{\text{new}}[0] = \mathbf{v}[0]$.
- (d) Take a rectangular grid with 5×6 grid points and a numbering of them $x_1, x_2, \dots, x_{30}$. Construct a matrix A such that $A[i, j] = \begin{cases} -4 & \text{if } x_i = x_j, \\ 1 & \text{if } x_i, x_j \text{ are vertically or horizontally neighbors,} \\ 0 & \text{otherwise.} \end{cases}$ You may use the concept of Kronecker product.